Portfolio RiskQ

Fast Tail Risk Estimation via Discretized IQAE

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Introduction

Risk teams need reliable VaR/CVaR under tight compute budgets, especially when tail behavior drives capital decisions. We present a classical-to-quantum pipeline that preserves the same loss distribution, tail threshold ℓ, and histogram while enabling IQAE-based tail probability estimation. The goal is a faster tail estimate with clear, auditable outputs that remain interpretable to stakeholders. Our workflow keeps the classical baseline in place while adding a quantum estimator side-by-side, so performance can be compared. This keeps results practical for risk review.

Background

- VaR: a loss cutoff; CVaR: the average of losses worse than that cutoff.
- Tail threshold ℓ: the line that marks what we call "extreme loss."
- Portfolio loss: $L = -w^\top r$; tail probability: $p = \mathbb{P}(L \geq \ell)$.
- Discretization: group losses into 2^n bins to make a simple histogram.
- IQAE: a quantum routine that estimates the tail probability from that histogram.
- Using the same ℓ and histogram keeps comparisons fair.

Objectives

- Build a working classical risk engine and dashboard MVP.
- Add discretization and tail-mask generation.
- Deliver a small-scale IQAE simulator with resource reporting.
- Communicate results through stakeholder-friendly visuals.

The emphasis is on a complete, inspectable pipeline that supports both classical validation and quantum-oriented estimation.

Models & Materials

Implementation uses Python with NumPy and Pandas for numerical work, a FastAPI backend for serving risk metrics, and a Streamlit + Plotly front end for interactive visualization.

- Gaussian return model; optional crash-mixture scenario.
- Nested Monte Carlo for regime uncertainty.
- Histogram-based discretization for quantum mapping.
- Backend toggle: classical vs quantum comparison path.

Methodology

Nested Monte Carlo (loss samples)

Draw returns from a Gaussian model, then compute portfolio loss as the weighted negative return. r are simulated asset returns, μ and Σ are their mean and covariance, and w are portfolio weights.

$$r \sim \mathcal{N}(\mu, \Sigma), \quad L = -w^\top r$$

Tail metrics

VaR is the loss quantile at level α; CVaR averages losses beyond VaR.

$$\mathrm{VaR}_\alpha = Q_\alpha(L)$$
$$\mathrm{CVaR}_\alpha = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha]$$

Discretization

- Bin losses into 2^n intervals.
- Produce edges, probabilities, tail mask.
- Bootstrap error estimate for bin stability.
- Bin counts n_i define histogram probabilities p_i; tail probability is the sum over tail bins.

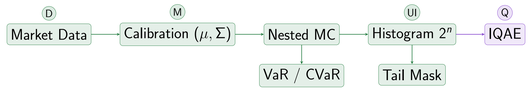
$$p_i = \frac{n_i}{N}, \quad \sum_i p_i = 1, \quad \hat{p} = \sum_{i \in \text{tail}} p_i$$

IQAE (simulator)

- Uses Grover iteration schedule to amplify tail states.
- Returns estimate with confidence interval.
- IQAE probes multiple Grover powers m to infer p from amplified measurements.

$$p = \sin^2(\theta), \quad p_m = \sin^2((2m+1)\theta)$$

System Pipeline (Schematic)



Market data calibrates (μ, Σ), nested MC yields losses, and discretization builds a 2^n histogram and tail mask in the classical pipeline. IQAE then estimates tail probability in the quantum stage.

Analysis & Results

- Validated probability normalization and tail-mask correctness.
- Compared IQAE estimates to true tail probability on toy data.
- Tracked resource usage (qubits, depth, 2Q gates, oracle calls).
- Compared classical vs quantum using the same ℓ and histogram.
- Dashboard showed VaR/CVaR and tail-highlighted histogram.

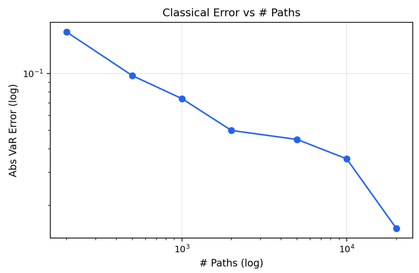
Key Contribution

We present a classical-to-quantum VaR/CVaR workflow that uses a discretized histogram and tail mask to enable IQAE comparison against nested Monte Carlo. We also map feasibility boundaries for NISQ constraints and validate the pipeline in a production-style dashboard. This bridges research-grade quantum estimation with production-style risk reporting, including confidence intervals and error metrics.

Experiment Visuals

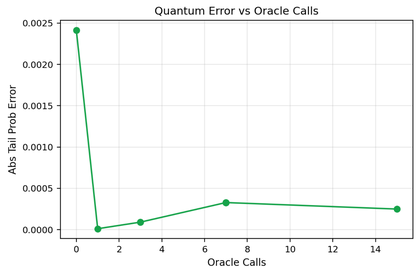
These plots summarize accuracy tradeoffs across classical sampling, quantum query cost, discretization resolution, and noise sensitivity. We report absolute tail-probability error against a known baseline, highlighting where additional compute yields diminishing returns and where noise dominates estimation error. The classical curve provides a sanity check on $1/\sqrt{N}$ behavior, while the IQAE curve shows how a few Grover rounds can capture most of the tail probability. Discretization and noise sweeps show the practical regime where IQAE comparisons remain stable.

Classical Error vs MC Paths



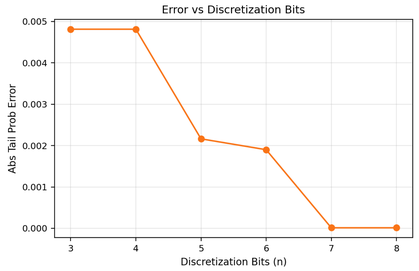
Takeaway: *VaR error falls as $1/\sqrt{N}$; 10k+ paths stabilizes tails.*

Quantum Error vs Oracle Calls



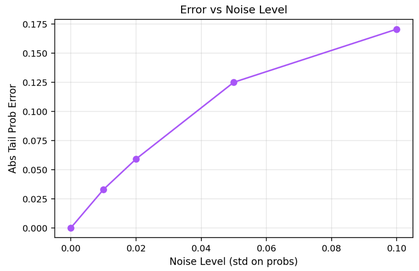
Takeaway: *Sub-1% tail error in a few Grover rounds.*

Error vs Discretization Bits



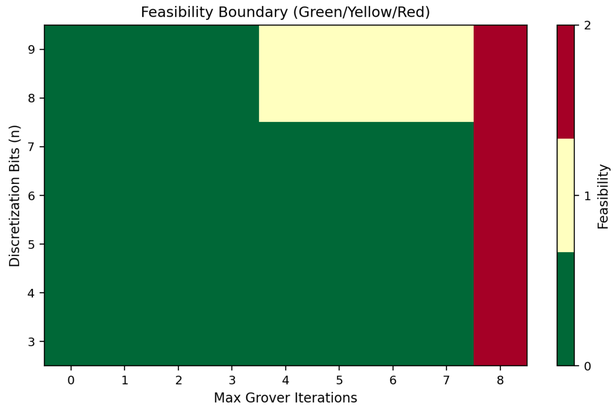
Takeaway: *Gains taper beyond 6–7 bits (binning bias).*

Error vs Noise Level



Takeaway: *Noise ($\sigma \gtrsim 0.05$) dominates error.*

Feasibility Boundary

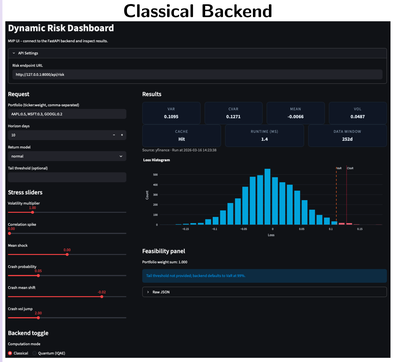


In the feasible region, resource demands stay low enough for near-term devices, while the borderline region signals settings that quickly become impractical. The not-viable region reflects NISQ-era limits where IQAE cost dominates the workflow. Bands use a composite resource score (qubits + depth + oracle calls).

Dashboard Visuals

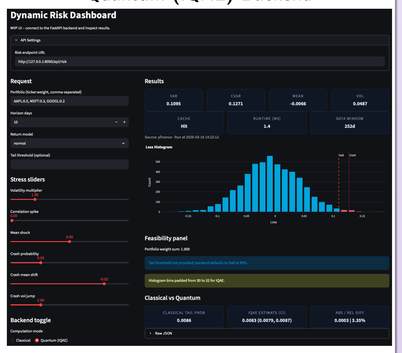
Left panel collects portfolio inputs and stress settings; right panel reports VaR/CVaR, a tail-highlighted histogram, and a classical vs quantum comparison. IQAE reports the same tail probability with a confidence interval and absolute/relative error.

Classical Backend



Interpretability: tail-highlighted histogram with VaR/CVaR markers.

Quantum (IQAE) Backend



Usability: IQAE CI + absolute/relative error vs classical.

Discussion & Implications

- Discretization provides a stable bridge to IQAE.
- Simulator estimates align with expected tail probabilities.
- Resource reports clarify near-term feasibility limits.
- Comparison uses the same threshold ℓ, scenario, and histogram.

Conclusion

The MVP validates a classical-to-quantum risk pipeline with consistent inputs, interpretable outputs, and a clear audit trail from data to tail estimate. IQAE offers a potential quadratic reduction in sampling complexity; our results focus on comparability and feasibility under near-term constraints.

References

Rockafellar & Uryasev (2000) CVaR, *Journal of Risk*; Jorion (2007) *Value at Risk*; Glasserman (2004) *Monte Carlo Methods in Financial Engineering*. Brassard et al. (2002) QAE, *Contemporary Mathematics*; Grinko et al. (2021) Iterative QAE, *npj Quantum Information*; Grover (1996) STOC; Woerner & Egger (2019) quantum risk analysis, *npj Quantum Information*; Amiet & Wehner (2020) NISQ finance algorithms, *npj Quantum Information*.

Acknowledgements & Contact

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https://github.com/amirorfali/Dynamic-Risk-Dashboard